Srujan Pendurthi

602-326-4325 | srujan.pendurthi@gmail.com | LinkedIn | Github| srujanpendurthi.com

Education

Carnegie Mellon University Pittsburgh, PA
Bachelor of Science in Artificial Intelligence, Additional Major in Business *Expected May 2028*

Experience

AI Engineering Intern May 2026 – Aug. 2026
Humana *Louisville, KY*

- Built multimodal RAG pipelines with Google ADK, streamlining Medical Assistant workflows by 33%
- Deployed LangChain and LangGraph agents into production workflows, increasing patient throughput by 6%
- Connected patient data and charts into the Athena EMR using FastAPI and a Snowflake MCP server
- Presented the agentic system in a final showcase to senior leadership and stakeholders

AI/ML Intern May 2025 – Aug. 2025
Drippy.ai

- Developed NLP and computer vision models that improved personalized recommendations by 28%
- Deployed and optimized scalable model inference pipelines, increasing platform engagement by 18%

Machine Learning Research Assistant May 2024 – July 2024
Grand Canyon University *Phoenix, AZ*

- Built a real-time ASL gesture translation system using OpenCV, MediaPipe, and PyTorch
- Trained an RNN classifier on EEG data for BCI-focused multimodal classification, improving accuracy by 25%
- Authored a 7-page paper and presented at the Arizona Association of the Deaf State Conference

Projects

whis — Local Study Engine | *Rust, Tauri v2, MLX, LanceDB, React* 2026

- Built a privacy-first local NotebookLM alternative for Apple Silicon: Rust/Tauri core, embedded LanceDB vector store, and MLX inference behind an OpenAI/Anthropic-compatible local server
- Implemented an agentic study engine with command states (`/plan`, `/deep-research`), Socratic questioning, and flashcard generation synced to Anki via AnkiConnect
- Ran a two-tier model setup (primary reasoning plus auxiliary routing, 4-bit/8-bit MLX) fully on-device with no cloud dependency

KV-Cache Retention Study | *Python, Transformers, CUDA/MPS* 2026

- Measured LLM KV-cache compression by distance from query across 340 trials and 3 models, showing token-scoring heuristics never recover mid-context (0 hits vs. uncompressed 100%)
- Proved via an oracle-retention control that the failure is observability not model capacity; wrote the results up as a paper

matalu | *Rust, Python, Apple MLX, QLoRA* 2026

- Built local real-time dictation chaining Parakeet TDT speech-to-text into a QLoRA-tuned Qwen2.5 cleanup model, orchestrated as supervised sidecars from a Rust core
- Added a guard that rejects hallucinated model output and falls back to the raw transcript
- Implemented Speculative Decoding and KV Caching for inference

Technical Skills

Languages: Python, Go, C/C++, TypeScript, SQL, Rust
AI/ML: LangChain, LangGraph, Google ADK, Claude SDK, PyTorch, Transformers, Apple MLX
Concepts: RAG, LLM agents, MCP, vector databases (LanceDB), prompt engineering, FastAPI
Developer Tools: Git, Docker, Claude Code, Codex, Cursor, VS Code, Jenkins, Linux, GCP, AWS

Fully work authorized. No visa sponsorship required. US Citizen.